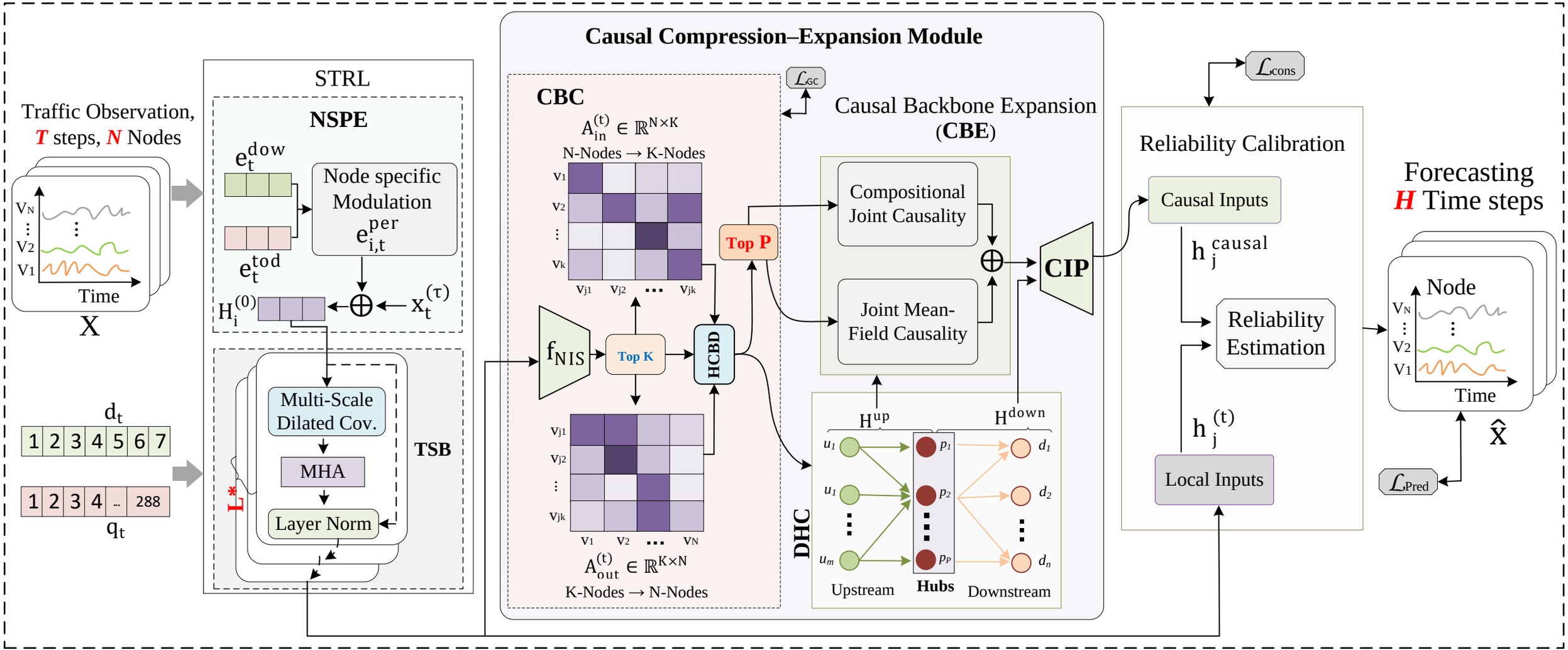

Traffic Observation,
T steps, N Nodes
X
STRL
NSPE
Node specific Modulation
Multi-Scale Dilated Cov.
MHA
Layer Norm
TSB
L*
Causal Compression–Expansion Module
CBC
N-Nodes → K-Nodes
K-Nodes → N-Nodes
Top K
Top P
HCBD
Causal Backbone Expansion (CBE)
Compositional Joint Causality
Joint Mean-Field Causality
CIP
DHC
Upstream
Hubs
Downstream
Reliability Calibration
Causal Inputs
Reliability Estimation
Local Inputs
Forecasting H Time steps
Node
Time